Course Description and Outcome Form
Department of Computer Science and Engineering
School of Engineering and Computer Science
BRAC University

A. Course General Information:

Course Code:	CSE 470
Course Title:	Software Engineering
Credit Hours (Theory):	3
Contact Hours (Theory):	3
Category:	Program Core
Type:	Lecture
Prerequisites:	CSE370
Co-requisites:	None

B. Course Catalog Description (Content):

Introduces students to all aspects of SDLC. Make them familiar with standard practices and techniques in the software industry. It aims to reduce the gap between academic and industry practices. Projects by students where they analyze the requirements and develop a system with industry best practices.

C. Course Objective

a. Introduce Students to fundamental concepts of SDLC
b. Teach students about the requirement analysis of the system
c. Teach students about how software engineers work in industry and the rules they follow.
d. Explain how to build software using different design patterns.
e. Teach students about managing programming processes and documentation.
f. Help students build a system using proper and modern tools.
g. Explain different software-building processes and methodologies, and refactoring.

D. Course Outcomes (COs):

Upon successful completion of this course, students will be able to

Sl.	CO Description	Weightage (%)
CO1	**Explain** the fundamental concepts of Software Engineering	20
CO2	**Analyze** requirements and design a high-level overview of the software	15
CO3	**Apply** different software architecture and patterns, including MVC	30
CO4	**Evaluate** the viability of software using testing and software metrics	15
CO5	**Apply** project management principles and economic feasibility analysis to plan and design a cost-effective software solution.	10
CO6	**Demonstrate** the ability to independently learn and apply modern tools, technologies, or methodologies in software development.	10

E. Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
CO1	**Explain** the fundamental concepts of Software Engineering	a	Cognitive/Analyze	Lectures, Notes, Project	Exam
CO2	**Analyze** requirements and design a high-level overview of the software	c	Cognitive/Create	Lectures, Notes, Project	Exam, Assignment, Project
CO3	**Apply** different software architecture and patterns, including MVC	d	Cognitive/Analyze	Lectures, Notes, Project	Exam, Assignment, Project
CO4	**Evaluate** the viability of software using testing and software metrics	i	Cognitive/Create	Lectures, Notes, Project	Exam
CO5	**Apply** project management principles and economic feasibility analysis to plan and design a cost-effective software solution.	k	Cognitive/ Apply	Discussion, Project	Exam, Project
CO6	**Demonstrate** the ability to independently learn and apply modern tools, technologies, or methodologies in software development.	l	Psychomotor/ Precision	Discussion, Project	Project

F. Course Materials:

i. Text and Reference Books:

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1	Clean Code: A Handbook of Agile Software Craftsmanship	Robert C. Martin	2017	–	PHI	10: 9780132350884
2	Design Patterns: Elements of Reusable Object-Oriented Software	Erich Gamma, Richard Helm, Ralph Johnson and John Vlissides	1994	–	Addison Wesley	10: 0201633612
3	Software Engineering: A Practitioner's Approach	Roger S. Pressman and Bruce Maxim	2020	9th	McGraw Hill	978-1-260-54800-6

ii. Other materials (if any)

Lecture Notes and presentation slides

iii. Lesson Plan: **(red-marked contents are not included in this semester)**

No	Topic	Week/Lecture	Related CO (if any)
1	Introduction to Software Engineering and software process model, different types of SDLC: Waterfall, V model, Incremental, Iterative, Spiral. Agile Development process and different frameworks (XP, AUP, Scrum)	**Week 1, 2**	CO1
2	Requirement Engineering- functional and non-functional, UML design, e.g., CLASS diagram	**Week 3, 4**	CO2
3	Introduction to Software Architecture and Design. Different types of software Architectural patterns: MVC, Layered, Repository, Client-Server, Microservices, Pipe-Filter, Github	**Week 5, 6**	CO3
Mid Exam			
4	Design Pattern and its type: Observer, Adapter, Factory and Singleton	**Week 7, 8**	CO3
5	Software Testing. Software quality metric (CFG, Cyclomatic Complexity, SIX)	**Weeks 9, 10**	CO4

6	Refactoring of Software Code	**Weeks 11**	CO5
7	Project Demonstration	**Week 12, 13**	CO2, CO3, CO5, CO6
Final Exam			

G. Assessment Tools:

Assessment Tools	**Weightage (%)**
1. Assignment	5%
2. Quizzes	15%
3. Mid-Term Examination	25%
4. Project	15%
5. Final	40%

H. CO Assessment Plan:

Assessment Tools						
	CO1	**CO2**	**CO3**	**CO4**	**CO5**	**CO6**
Homework						
Quizzes						
Examinations	x	x	x	x	x	
Project						x

I. CO Attainment Policy:

As per the Department of CSE course outcome attainment policy

J. Grading policy:

As per the BRAC University grading policy

K. Course Coordinator:

Atanu Roy (ART)